# Research Statement

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## Overview

I am an applied economist working on energy and environmental economics, international trade and the environment, and the empirical methods used to study them. One question runs through my research: how can emerging and developing economies keep growing while reducing the environmental cost of that growth? I study this question through the channels that policy can influence: energy structure, financial development, technology and institutions, and the structure of what countries produce and trade.

My approach is empirical and comparative. I work with long national time series, cross-country and provincial panels, spatial models, trade networks and machine-learning tools, and I choose the method to fit the question and the data. Much of my work concerns South Asia, China and other large emerging economies (the BICS group of Brazil, India, China and South Africa, and the MERCOSUR bloc), where the trade-off between development and environmental quality is most acute and where empirical evidence is still uneven.

My training shaped this agenda. I studied agricultural and resource economics at the University of Agriculture, Faisalabad, and econometrics at the Pakistan Institute of Development Economics. I then completed a PhD in World Economy at Zhongnan University of Economics and Law, where my dissertation applied machine-learning methods and the gravity model to identify robust determinants of bilateral trade. Postdoctoral positions at Hainan University (2020–2022) and Guangzhou University (2022–2024) allowed me to build the research programme described below. Since January 2025 I have been an Associate Professor of Economics at Quanzhou University of Information Engineering. To date, my work comprises sixteen peer-reviewed journal articles, one conference proceedings paper and four book chapters.

My research is organised into three connected strands: (1) finance, energy and environmental quality in emerging economies; (2) trade, economic complexity and the environment; and (3) machine learning, technology and sustainable production. A methodological thread links them, and a provincially funded project on industrial upgrading carries the agenda forward.

## 1. Finance, Energy and Environmental Quality in Emerging Economies

**Question.** Does financial deepening help or hinder decarbonisation, and how do energy use, natural-resource dependence, digital technology and institutions shape environmental outcomes in fast-growing economies? The finance–environment literature gives mixed answers, and much of it uses carbon emissions as the only measure of environmental quality.

**India as a case study.** My first line of work examines India, one of the world's largest emitters and a country with a rapidly developing financial sector. In Qayyum et al. (2021, *Energies*), my coauthors and I study financial development, renewable energy consumption, technological innovation and $CO_2$ emissions over 1980–2019. We use the autoregressive distributed lag (ARDL) bounds approach for short- and long-run dynamics and a vector error-correction model for causal direction, and we check robustness with FMOLS, DOLS and canonical cointegrating regression. Financial development raises emissions, while renewable energy and technological innovation reduce them in both the short and the long run. Economic growth and urbanisation add to environmental pressure. A companion study, Qayyum et al. (2022, *Energy & Environment*), turns from the level of financial development to its *stability*. Using data for 1980–2020, it finds that financial instability has no significant effect on India's emissions once energy use, income and

urbanisation are controlled for, and it supports an environmental Kuznets curve. Read together, the two papers suggest that the environmental footprint of finance in India works mainly through the scale and direction of credit, not through volatility, a distinction that matters for green-finance policy.

**Cross-country evidence.** In Yang, Ali, Nazir, Ullah and Qayyum (2020, *Air Quality, Atmosphere & Health*), we take the same question to a panel of 54 developing economies over 1980–2016, using fixed-effects and difference and system GMM estimators. In this broader sample, financial instability is associated with lower emissions, while growth, energy consumption and urban population increase them. The contrast with the Indian results shows why single-country and panel evidence should be read together.

**Beyond carbon: ecological footprint and load capacity.** More recent work broadens the measure of environmental quality. In Qayyum, Zhang, Ali and Kirikkaleli (2024, *Environmental Technology & Innovation*), we study how information and communication technology (ICT) and institutional quality affect the ecological footprint of the MERCOSUR economies over 1990–2021. Because these economies are closely integrated, we use second-generation panel methods that allow for cross-sectional dependence: Westerlund cointegration tests and the cross-sectionally augmented ARDL (CS-ARDL) estimator, with the augmented mean group estimator as a robustness check. ICT and institutional quality each reduce the ecological footprint, and their interaction reduces it further. Trade, energy use and growth enlarge the footprint. In Qayyum, Bashir, Ali and Mehmood (2026, *Energy & Environment*), we use the load capacity factor, which compares an economy's biocapacity with its ecological footprint, alongside $CO_2$ emissions to study natural-resource depletion in the BICS economies over 1970–2021. Pooled mean group estimation shows a sign reversal: natural-resource abundance improves environmental quality in the short run but increases pollution in the long run. Primary energy use, urbanisation and growth weaken environmental quality throughout.

**Contribution.** This strand provides country- and bloc-level evidence on how financial, energy and institutional conditions shape environmental outcomes, and shows that conclusions can change with the measure of environmental quality and the time horizon. The policy message is consistent: renewable energy, technological upgrading and institutional strength are the margins on which emerging economies can separate growth from environmental damage.

## 2. Trade, Economic Complexity and the Environment

**Question.** How do the composition and network structure of international trade, and the productive capabilities they reflect, shape carbon emissions, green innovation and structural transformation? This strand builds directly on my doctoral training in trade.

**Economic complexity and embodied carbon.** In Qayyum, Yu and Li (2021, *Environmental Science and Pollution Research*), we ask whether more sophisticated production structures change the carbon embodied in trade, which matters for assigning responsibility for emissions across borders. Using dynamic panel GMM for 34 OECD and 24 non-OECD countries over 1995–2015, we find that economic complexity affects carbon embodied in exports and imports differently. Higher complexity also weakens the marginal effect of export scale and export structure on the foreign emissions embodied in exports.

**What drives complexity.** Two papers study the determinants of economic complexity. In Yu and Qayyum (2021, *International Journal of Finance & Economics*), we use a panel of 120 economies over 1996–2016 and find that financial openness, measured both de facto and de jure, raises economic complexity. For low- and middle-income countries, participation in global value chains and lower banking concentration also matter. In Zhang, Qayyum and Yu (2023, *Journal of Cleaner Production*), we use spatial panel models for 123 countries over 1995–2018 to separate the local and spillover effects of resource rents on complexity. Overall resource abundance lowers local complexity, consistent with a resource curse. The effect differs by resource: oil and mineral rents reduce complexity, while gas rents are associated with a "resource blessing". The spillovers also differ, being negative for oil and gas rents and positive for coal and mineral rents. Institutional quality and financial openness soften the adverse effects, while larger FDI inflows can strengthen them.

**Green trade networks and innovation.** In Yu, Qayyum and Li (2024, *Energy Economics*), we build a directed trade network of environmental goods, such as renewable-energy plant and wastewater-management equipment, covering 234 countries and regions over 2002–2021. We use network topology to trace how the network has evolved. The network has become more interconnected, and it is led by several OECD economies (France, Germany and the United States) and by China, India and South Africa. Quantile regressions for 27 OECD and 5 BRICS countries show that a more central position in the network promotes green technological innovation across the distribution. Betweenness and closeness centrality have non-linear, U-shaped effects. Environmental policy stringency supports green innovation in OECD countries, but non-market-based instruments in the BRICS may make it harder. The paper shows that a country's *position* in trade networks, and not only its trade volume, shapes the diffusion of clean technology.

**Openness, culture and innovation.** Related papers widen the view of trade. Using system GMM and path analysis on 31 Chinese provinces, Qayyum et al. (2022, *PLoS ONE*) finds that trade openness and stronger intellectual-property protection promote regional innovation, with the effect concentrated in the eastern provinces. Two papers study imported cultural goods as a less obvious channel of environmental change. A panel VAR for 158 economies (Yu and Qayyum, 2022, *Environmental Science and Pollution Research*) finds that these imports reduce emissions, and a GMM study of 121 countries (Qayyum, Li, Jian and Yu, 2023, *Energy & Environment*) shows that the effect is negative in higher-income countries but positive in lower-income ones.

**Contribution.** This strand links trade composition, network position and productive capabilities to environmental and innovation outcomes. It also brings spatial and network methods into questions that are usually studied with country-by-country regressions.

## 3. Machine Learning, Technology and Sustainable Production

**Question.** Can data-driven methods give more robust answers than conventional specifications to questions about energy, emissions and productivity, especially when there are many candidate variables, the relationships are non-linear, or prediction is itself a policy goal?

My interest in machine learning began with my dissertation, which used machine-learning methods together with the gravity model to separate robust from fragile determinants of bilateral trade. I have since applied these tools to environmental questions. In Qayyum et al. (2023, *Journal of Environmental Management*), we study technology, innovation and greenhouse-gas emissions in agriculture across 21 agriculture-dependent Asian economies. We use LASSO and Elastic-Net regressions to select among many candidate drivers and compare ASEAN, SAARC and GCC countries. ASEAN economies emerge as active adopters of technologies that raise output while protecting environmental quality. In SAARC economies, weaker adoption coincides with more volatile environmental quality, and agricultural $N_2O$ and $CO_2$ emissions rise with output. In GCC economies, there is little evidence of technology adoption. Across groups, deforestation and cropland expansion raise output but also raise agricultural emissions. I extended this work as Principal Investigator of a Guangzhou postdoctoral research project (2024) on how artificial-intelligence development and economic growth relate to sustainable agriculture in Asian countries.

In Qayyum, Ali, Hassan and Mehmood (2026, *Environment, Development and Sustainability*), we bring machine-learning regression and prediction together with the ARDL framework to study the energy–economy–environment trilemma in Pakistan over 1965–2022. The ARDL results show that energy use raises $CO_2$ emissions and that renewable energy reduces them only in the short run. The machine-learning projections indicate that emissions will rise again from 2024 as non-renewable energy use grows. The two sets of results support the same policy conclusion: a lasting reduction in emissions requires a structural shift from fossil fuels to renewables, not marginal adjustments.

Within this strand I have also worked on efficiency measurement and decision analysis. With colleagues at Hainan University, I coauthored a comparison of service-sector energy efficiency across Chinese provinces under heterogeneous resource endowments, using an undesirable-output super-efficiency SBM model with a Malmquist–Luenberger index (Sun, Liu and Qayyum, 2021, *Journal of Cleaner Production*). The paper received a Second Prize in the 12th Hainan Provincial Awards for Outstanding Social Science Achievements, with its first author, Peng Sun, as the lead recipient. I later took part in a National Natural Science

Foundation of China project on total-factor energy efficiency (No. 72164008, 2022–2025). With coauthors, I also evaluated the Belt and Road Initiative by combining SWOT analysis with weighted AHP and entropy methods (Qayyum et al., 2023, *Multimedia Tools and Applications*). Coauthored book chapters on machine learning in urban ecology and on carbon-emission patterns reflect my interest in combining economic and geospatial data.

**Contribution.** This strand shows how machine learning can complement econometrics in environmental economics: it helps with variable selection and prediction, while econometric models supply interpretable short- and long-run parameters.

## Methodological Contributions

Across these strands I have built a methodological toolkit:

- **Time-series cointegration and causality:** ARDL bounds testing, error-correction and VECM causality, with FMOLS, DOLS and CCR robustness checks, for long single-country series.
- **Heterogeneous and cross-sectionally dependent panels:** pooled mean group, CS-ARDL and augmented mean group estimators, and second-generation cointegration tests, for groups of closely integrated economies.
- **Dynamic panels and panel VAR:** difference and system GMM, and panel VAR with impulse responses and variance decompositions.
- **Spatial and network econometrics:** spatial panel models that separate direct and spillover effects, and complex-network measures of trade centrality combined with quantile regression.
- **Machine learning:** penalised regression (LASSO, Elastic-Net) for variable selection, and machine-learning regression and prediction alongside econometric benchmarks.
- **Efficiency and decision analysis:** SBM data envelopment analysis with undesirable outputs, Malmquist–Luenberger productivity indices, and multi-criteria methods (AHP, entropy weighting).

I work in R and Stata, with Python and ArcGIS for machine-learning and spatial tasks. My guiding principle is triangulation: where possible I confirm results with an alternative estimator, so that no policy conclusion rests on a single specification.

## Policy Relevance

My research speaks to policy questions faced by emerging economies. The India studies suggest that green-finance policy should steer credit toward renewables and innovation, and the Pakistan study indicates that the shift to renewable energy must be structural rather than marginal if emissions are to fall durably. The MERCOSUR and BICS studies point to institutions, digital infrastructure and natural-resource governance as levers for environmental quality. The trade and complexity work shows that a country's network position and productive capabilities shape its access to clean technology, which bears on environmental-goods liberalisation and industrial policy. The agricultural study shows that technology adoption can reconcile food production with lower emissions in Asia, but that the gains vary widely across regions.

## Ongoing Work

My main current project, which I lead as Principal Investigator, is funded by the Fujian Provincial Department of Education (No. JAT251221, 2026–2028): *"Strategic Choices for Industrial Upgrading in Fujian Province: Opportunities in the China–Pakistan Economic Corridor Based on Product Space."* It uses product-space and economic-complexity methods to identify products in which Fujian has latent comparative advantage and to assess which of them match demand and capabilities along the China–Pakistan Economic Corridor. The project brings together my work on economic complexity and on South Asian development, and aims to produce evidence that provincial policymakers can use.

## Future Research Agenda (Next Three to Five Years)

I plan to extend my agenda in four directions.

1. **Asymmetric and non-linear environmental dynamics.** Many energy–environment relationships are unlikely to be symmetric: a rise in renewable-energy use or in financial development need not have the same effect as a fall of the same size. I plan to use asymmetric frameworks, including nonlinear ARDL and its panel extensions, together with quantile-based methods, to test for such asymmetries across income groups and regions.
2. **Interpretable and causal machine learning.** I aim to move from machine learning as a prediction tool toward methods that support causal and interpretable inference, such as double/debiased machine learning and model-explanation techniques. The goal is to estimate heterogeneous policy effects in energy and environmental settings while keeping the transparency that policy audiences need.
3. **Product space, green capabilities and industrial upgrading.** Building on the Fujian project, I plan to study how regions and countries can move into green and higher-complexity products, how trade-network position affects the diffusion of clean technology, and what this implies for corridor-based cooperation such as the China–Pakistan Economic Corridor.
4. **Technology, digitalisation and sustainable agriculture.** Continuing my postdoctoral work, I intend to examine how digital and AI-related technologies affect agricultural productivity and emissions, and to combine economic data with geospatial and Earth-observation data where possible.

I expect these projects to lead to journal articles and to applications for national and provincial research funding, and to deepen my collaborations with coauthors in China and abroad.

## Research, Teaching and Student Supervision

My research feeds directly into my teaching. In *International Trade Theory and Practice*, I use my work on trade networks, economic complexity and environmental goods to show students how trade patterns are measured and why they matter for development and the environment. In *Macroeconomics*, the energy–growth–emissions evidence provides concrete cases for discussing growth, structural change and policy trade-offs. In *Management Studies*, I draw on the policy side of my research. I emphasise hands-on empirical skills, with the aim that students learn to estimate, test and interpret models on real public data in R or Stata.

In supervising students, I aim to turn these skills into research. My projects offer manageable and well-defined topics for undergraduate theses and early graduate research: replicating and extending a cointegration analysis for a new country, mapping a product-space or trade network, or comparing machine-learning and econometric forecasts. I encourage students to frame a clear policy question and to check that their results hold under alternative methods. As a peer reviewer for more than fifteen international journals, I also try to pass on the standards of careful empirical work that I apply to my own research.

## Selected Publications

Full list: personal website · Google Scholar · ORCID. * = corresponding author.


1. **Qayyum, M.**, Bashir, M. A., Ali, M., & Mehmood, U. (2026). A path to carbon neutrality: Exploring the impact of natural resource use on CO2 emissions and load capacity factor in BICS. *Energy & Environment*. Advance online publication. https://doi.org/10.1177/0958305X261474724
2. **Qayyum, M.**, Ali, M., Hassan, M. W., et al. (2026). Harnessing machine learning to decode the energy–economy–environment trilemma: Insights from renewable and non-renewable energy use in Pakistan. *Environment, Development and Sustainability*. Advance online publication. https://doi.org/10.1007/s10668-026-07985-z
3. Yu, Y., **Qayyum, M.***, & Li, S. (2024). Trade dynamics of environmental goods within global energy economy and their impacts on green technological innovation: A complex network analysis. *Energy Economics*, 107957. https://doi.org/10.1016/j.eneco.2024.107957
4. **Qayyum, M.**, Zhang, Y., Ali, M., & Kirikkaleli, D. (2024). Towards environmental sustainability: The role of information and communication technology and institutional quality on ecological footprint in MERCOSUR nations. *Environmental Technology & Innovation*, 103523. https://doi.org/10.1016/j.eti.2023.103523
5. Zhang, Y., **Qayyum, M.***, & Yu, Y. (2023). Effects of resource abundance on economic complexity: Evidence from spatial panel model. *Journal of Cleaner Production*, 139134. https://doi.org/10.1016/j.jclepro.2023.139134

6. **Qayyum, M.**, Zhang, Y., Wang, M., Yu, Y., Li, S., Ahmad, W., Maodaa, S. N., Sayed, S. R. M., & Gan, J. (2023). Advancements in technology and innovation for sustainable agriculture: Understanding and mitigating greenhouse gas emissions from agricultural soils. *Journal of Environmental Management*, 347, 119147. https://doi.org/10.1016/j.jenvman.2023.119147
7. **Qayyum, M.**, Yu, Y., Nizamani, M. M., Raza, S., Ali, M., & Li, S. (2022). Financial instability and CO2 emissions in India: Evidence from ARDL bound testing approach. *Energy & Environment*. https://doi.org/10.1177/0958305X211065019
8. **Qayyum, M.**, Ali, M., Nizamani, M. M., Li, S., Yu, Y., & Jahanger, A. (2021). Nexus between financial development, renewable energy consumption, technological innovations and CO2 emissions: The case of India. *Energies*, 14(15), 4505. https://doi.org/10.3390/en14154505
9. **Qayyum, M.**, Yu, Y., & Li, S. (2021). The impact of economic complexity on embodied carbon emission in trade: New empirical evidence from cross-country panel data. *Environmental Science and Pollution Research*. https://doi.org/10.1007/s11356-021-14414-3
10. Yu, Y., & **Qayyum, M.*** (2021). Impacts of financial openness on economic complexity: Cross-country evidence. *International Journal of Finance & Economics*. https://doi.org/10.1002/ijfe.2491
11. Sun, P., Liu, L., & **Qayyum, M.** (2021). Energy efficiency comparison amongst service industry in Chinese provinces from the perspective of heterogeneous resource endowment: Analysis using undesirable super efficiency SBM-ML model. *Journal of Cleaner Production*, 328, 129535. https://doi.org/10.1016/j.jclepro.2021.129535 *Second Prize, 12th Hainan Provincial Awards for Outstanding Social Science Achievements (first author as lead recipient).*

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